

VLM Model: GPT-4o



Physical Reasoning Query Prompt:

Please infer the moveable object's physics properties and confidence values according to the **visual input** and **textual instruction**.



Instruction Optimization Query Prompt:

You are a skilled assistant for **refining text description**. Your goal is to take the inputs and generate an optimized description for video generation: **1)** Refine the draft instruction to **align with the motion dynamics** and physical properties. **2)** **Integrate low-confidence physical properties** by inferring missing details where necessary. **3)** Ensure the output is suitable for generating video.

Initial Physical Reasoning



image



P_object {value, conf}:

Density (ρ): {800, 0.7}
Mass (m): {0.5, 0.8}
YoungsModulus (E): {1e5, 0.5}
PoissonsRatio (ν): {0.4, 0.5}

...
low-conf sets: { E, ν } ($\text{conf} < \gamma$)

Instruction: a horse toy standing on a toy green bird is spinning.

Motion Probe Generation and Iterative Reasoning

Instruction



I2V



probe video



P_object {value, conf}:

Density (ρ): {800, 0.7}
Mass (m): {0.5, 0.8}
YoungsModulus (E): {1e6, 0.6}
PoissonsRatio (ν): {0.4, 0.5}

...
low-conf sets: { ν } ($\text{conf} < \gamma$)



Optimized Instruction



I2V



probe video



P_object {value, conf}:

Density (ρ): {800, 0.7}
Mass (m): {0.5, 0.8}
YoungsModulus (E): {1e6, 0.6}
PoissonsRatio (ν): {0.35, 0.7}

...
low-conf sets: None